

EC442: Lecture Notes

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Notes on the 1st year Macro class

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1. Lecture 1 Sep. 29

1.1. Why heterogeneity

Complete markets + standard welfare assumptions:

- Competitive eq. Pareto efficient; planner representation w/ fixed weights $\alpha_i > 0$.
- Common discounting; preferences $u(c) - v(n)$. Consumption FOCs:

$$\alpha_i u'(c_t^i) = \alpha_j u'(c_t^j) \Rightarrow \frac{u'(c_t^i)}{u'(c_t^j)} = \frac{\alpha_j}{\alpha_i}. \quad (1.1)$$

Marginal-utility ratio constant across time/states.

- Common CRRA: $u'(c) = c^{-\sigma}$.

$$\frac{c_t^i}{c_t^j} = \left(\frac{\alpha_i}{\alpha_j} \right)^{\frac{1}{\sigma}}. \quad (1.2)$$

Fixed consumption shares $\rightarrow$ **perfect insurance** against idiosyncratic income shocks.
Aggregate risk remains; unequal Pareto weights $\rightarrow$ unequal consumption.

- Strictly convex labor disutility; interior labor choice; aggregate conditions fixed: productivity $\uparrow \rightarrow$ labor $\uparrow$; consumption share unchanged.

Implications:

- Karl Marx
- Full insurance at odds w/ data

1.2. How to model incomplete markets?

1. **Exogenous:** assume some markets absent.
 1. e.g. save/borrow via noncontingent bonds; no state-contingent claims.
2. **Endogenous:** derive market incompleteness from frictions.
 1. e.g. limited commitment (participation constraints), private information.
 2. deeper but harder to do

1.3. Planning problem

Set $n_t = A = 1$; $f(k) = F(k, 1)$. Given $k_0 \geq 0$. Discount factor $0 < \beta < 1$; depreciation $0 \leq \delta \leq 1$.

Choose sequences $c_t, k_{t+1} \geq 0$:

$$\max \sum_{t=0}^{\infty} \beta^t u(c_t) \text{ s.t. } c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t. \quad (1.3)$$

- **Lagrangian**; current-value multipliers λ_t :

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t [u(c_t) + \lambda_t (f(k_t) + (1 - \delta)k_t - c_t - k_{t+1})]. \quad (1.4)$$

- **Interior FOCs:**

$$\begin{aligned} c_t : u'(c_t) &= \lambda_t, \\ k_{t+1} : \lambda_t &= \beta \lambda_{t+1} [f'(k_{t+1}) + 1 - \delta]. \end{aligned} \quad (1.5)$$

Euler equation:

$$u'(c_t) = \beta u'(c_{t+1}) [f'(k_{t+1}) + 1 - \delta]. \quad (1.6)$$

- **Transversality condition:**

$$\lim_{T \rightarrow \infty} \beta^T \lambda_T k_{T+1} = \lim_{T \rightarrow \infty} \beta^T u'(c_T) k_{T+1} = 0. \quad (1.7)$$

Discounted shadow value of terminal capital $\rightarrow 0$.

- Increasing, concave u ; concave f ; finite discounted utility: feasibility + FOCs + TVC $\rightarrow$ global optimum. At corners, use KKT conditions.

1.4. Towards a recursive formulation

- Sequence approach: infinitely many dated choices + optimality conditions.
- Repeated trade-off: consume today vs. save for tomorrow.
- Stationary problem; current capital k_t summarizes relevant history.
- Recursion: continuation value summarizes future choices; same decision problem at every date.

Value function: maximal lifetime utility from initial capital k_0 :

$$v^*(k_0) = \max_{\{c_t, k_{t+1}\}_{t \geq 0}} \sum_{t=0}^{\infty} \beta^t u(c_t). \quad (1.8)$$

Feasible sequences: $c_t, k_{t+1} \geq 0$; $c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t$, all $t \geq 0$.

Split date 0 from continuation:

$$v^*(k_0) = \max_{c_0, k_1} \left[u(c_0) + \beta \max_{\{c_t, k_{t+1}\}_{t \geq 1}} \sum_{t=1}^{\infty} \beta^{t-1} u(c_t) \right]. \quad (1.9)$$

- Outer max: $c_0, k_1 \geq 0$; $c_0 + k_1 = f(k_0) + (1 - \delta)k_0$.
- Inner max: given chosen k_1 , same feasibility constraints for all $t \geq 1$. Discounting restarts at date 1 $\rightarrow$ inner value $v^*(k_1)$.

Continuation value absorbs future choices. **Bellman equation** (current capital k , next-period capital k'):

$$v^*(k) = \max_{c, k'} [u(c) + \beta v^*(k')] \quad (1.10)$$

subject to

$$c + k' = f(k) + (1 - \delta)k, c, k' \geq 0. \quad (1.11)$$

Infinite sequence $\rightarrow$ one-period choice + discounted continuation value. Stationarity $\rightarrow$ same problem at every date.

Bellman equation = **functional equation**; unknown = function $v(k)$.

Questions:

- **Existence**: conditions for solution v ?
- **Uniqueness**: conditions for unique v within chosen function class?
- **Equivalence**: when does $v = v^*$, with maximizing policy solving original sequence problem?

Answer:

- *Recursive Macroeconomic Theory* (RMT), “Dynamic Programming” (Ljungqvist & Sargent, 2018, ch. 3).

1.5. Dynamic Programming

Bellman operator T : candidate continuation value $v \rightarrow$ updated value Tv .

$$(Tv)(k) = \max_{c,k'} [u(c) + \beta v(k')] \quad (1.12)$$

subject to

$$c + k' = f(k) + (1 - \delta)k, c, k' \geq 0. \quad (1.13)$$

Bellman solution = **fixed point**:

$$Tv = v \Leftrightarrow (Tv)(k) = v(k) \text{ for all feasible } k. \quad (1.14)$$

Fixed-point theorems give existence/uniqueness conditions within chosen function space; assumptions matter.

1.6. Contraction mapping theorem

Theorem 1.6.1 (Contraction mapping theorem): Let (S, d) be a nonempty complete metric space and $T : S \rightarrow S$ a contraction: there exists $\beta \in (0, 1)$ such that

$$d(Tx, Ty) \leq \beta d(x, y) \quad \text{for all } x, y \in S. \quad (1.15)$$

Then T has a unique fixed point $v \in S$. For every $v_0 \in S$ and $n = 0, 1, 2, \dots$,

$$d(T^n v_0, v) \leq \beta^n d(v_0, v). \quad (1.16)$$

Hence $T^n v_0 \rightarrow v$ as $n \rightarrow \infty$, with geometric convergence. Here T^n denotes n successive applications of T , with T^0 the identity.

1.6.1. Application to planner

- Assume feasible capital remains in nonempty compact interval K .
- $S = C(K)$: continuous real-valued functions on K ; automatically bounded.
- **Sup-norm metric**:

$$d(v, w) = \|v - w\|_\infty = \sup_{k \in K} |v(k) - w(k)|. \quad (1.17)$$

(S, d) complete.

- Must verify $T : S \rightarrow S$: updated values remain bounded and continuous. Then Bellman discounting gives

$$\|Tv - Tw\|_{\infty} \leq \beta \|v - w\|_{\infty}. \quad (1.18)$$

Theorem 1.6.1 $\rightarrow$ unique fixed point in S ; value iteration converges from any $v_0 \in S$.

1.7. Using the value function analytically

Assume u , f , and v differentiable; optimal choices interior. Substitute $c = f(k) + (1 - \delta)k - k'$ into Bellman:

$$v(k) = \max_{k'} [u(f(k) + (1 - \delta)k - k') + \beta v(k')], \quad (1.19)$$

over feasible k' .

- **FOC for k' :**

$$-u'(c) + \beta v'(k') = 0 \quad \Rightarrow \quad u'(c) = \beta v'(k'). \quad (1.20)$$

- **Envelope condition:** differentiate w.r.t. k , holding optimal k' fixed:

$$v'(k) = u'(c)[f'(k) + 1 - \delta]. \quad (1.21)$$

- **Combine:** apply envelope condition next period $\rightarrow$ usual Euler equation:

$$u'(c_t) = \beta v'(k_{t+1}) = \beta u'(c_{t+1})[f'(k_{t+1}) + 1 - \delta]. \quad (1.22)$$

Same condition as sequence problem. At corners, use KKT conditions.

Bibliography

Ljungqvist, L., & Sargent, T. J. (2018). *Recursive Macroeconomic Theory* (4th ed.). MIT Press. <https://mitpress.mit.edu/9780262038669/recursive-macroeconomic-theory/>^o